

Information Retrieval and Text-Mining

Guidelines Tutorials

Department of Advanced Computer Sciences (DACS)
Master of AI / DSDM
Course year 2025-2026

Background

- We have moved away from Google Colab
- We now use our own cloud solution which can be found at: www.irtm-course-um.nl
- The cloud servers run in Amsterdam, the Netherlands.

Student Login Instructions

Access the Hub: Go to <https://www.irtm-course-um.nl>.

Username: Log in using your **Maastricht University i-number** (e.g., i123456789).

Password: The **first password** you enter will become your permanent password for this system. Please choose a strong password and remember it carefully.

Coursework: Once logged in, click on the **Assignments** tab in the sidebar to view, fetch, and submit your notebooks for the *Information Retrieval and Text Mining* course. More information in the `README.md`, `LOCAL_SETUP.md` and `requirements.txt`.

The Cluster

- We have a pretty basic cluster with up to 40 VCPU's, 82 Gb RAM, 200 Gb SSD, an A100 with 16Gb, and up to 20 instances.
- We are with 20 students, so do not all wait until the last moment to run your tutorials!

Run on your own hardware

- You can also do the development on your own computer and then upload that file to the assignment folder on the cluster. See the additional instructions further on.
- The assignments are on GitHub:
<https://github.com/TextMiningUM/IRTM-2025-2026>



Assessment

- We use NBGrader for the (automatic) assessment where possible.
- Our Teaching Assistant will check the automatic assessments of the mandatory assignments and also assess those parts that cannot be done automatically.

Starting

- Go to www.irtm-course-um.nl and log in with your student number. This is your login name.
- First time login you can create your own password. For further instructions, use the references in the help menu from Jupyter.
- Open a terminal and run to get the tutorial files from GitHub:

```
git clone https://github.com/TextMiningUM/IRTM-2025-2026.git
```

The Tutorial Assignments

- Open the folder tool on the left. Navigate to the `Assignments/` folder. This is where all the notebooks are.
- We recommend you do all tutorials and learn from the questions (these are like the exam questions) and understand how to use the libraries for the mandatory tutorials.
- Complete the exercises and submit as instructed. You can also submit the non-mandatory tutorials for grading, so you know if your answers were correct.



GitHub

- When there are new assignments, you can get them using (does not overwrite your changes):

```
cd IRTM-2025-2026  
git pull
```

Or (to overwrite your changes)

```
cd ~/IRTM-2025-2026  
git checkout -- .  
git pull
```

Additional Tips

- We recommend using the working directory `Workspace` on your Jupyter notebook and copy the notebooks from the `Assignment` folder there when you start working. Once finished, you can copy them to `SubmittedWork` folder.
- **Sometimes, the notebook will not start.** In that case, use the option `Kernel – Restart Kernel`.
- Unlike Colab, there is no second counter when a cell runs. Running cells have `[*]` in front of them. Cells that are done have a number. If you run 'all cells', the cell running is the first with `[*]`. All others (that still need to be run), will also show `[*]`.

Running the Notebooks on your own hardware

See the file `requirements.txt` from Git: Contains all 40+ Python packages needed across all 11 tutorials, organized by category (core scientific stack, NLP, deep learning, search & retrieval, topic modelling, OpenAI, etc.). Matches Python 3.10 as used on your cluster (from the Dockerfile).

See the file: `LOCAL_SETUP.md`, a comprehensive student guide covering:

- Prerequisites — Python 3.10, 8–16 GB RAM, ~10 GB disk
- Environment setup — conda or venv instructions for Windows/macOS/Linux
- Dependency installation — single `pip install -r requirements.txt`
- Java JDK 11+ — needed only for Tutorial 03 (Pyserini/Lucene)
- NLTK data downloads — all 17 corpora/models in one command
- spaCy model — `en_core_web_sm` for Tutorial 10
- GPU support — which tutorials benefit (07 & 08 benefit most), how to install CUDA PyTorch
- OpenAI API key — for Tutorials 11 & 12, with cost estimate (~€1–€3)
- Tutorial-specific notes — e.g. Tutorial 07 must run before 11 (generates required data files)
- Submission workflow — reminding students to still submit via JupyterHub
- Troubleshooting — common errors and fixes

Additional remarks on notebooks:

Java is needed only for Tutorial 03 (Pyserini)

GPU is optional everywhere but strongly recommended for Tutorials 07 & 08 (BERT fine-tuning)

OpenAI API (paid) is required for Tutorials 11 & 12

Tutorials 07 → 11 have a data dependency (07 generates JSON files used by 11)

Mandatory tutorials

- 3 IRTM Search Engines
- 7 IRTM Building Structured Representations
- 11 IRTM Conversational Search Sticking to the Facts.

Submitting your work

- After a git clone, all tutorials are there:
`/IRTM-2025-2026/Assignments/`
- You can do your work on the Jupyter cluster or on your own computer.
- When you are done, copy your note book to the corresponding folder under: `/IRTM-2025-2026/SubmittedWork/`



Grading

- At the time communicated, we will (auto)grade your submissions and communicate the results.
- There will be feedback in the folder 'feedback' of the tutorial under SubmittedWork.
- You can also ask questions in the lectures specially dedicated to the tutorials.

